

TONGHE, China

WMO: 509630

Lat: 45.9825N

Lon: 128.7270E

Elev: 110

StdP: 100.01

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.7	-28.5	-33.5	0.2	-30.4	-31.5	0.2	-28.3	12.3	-15.1	11.2	-17.2	2.1	250	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	30.3	22.0	29.0	21.8	27.6	21.3	24.5	28.0	23.6	27.1	22.7	26.0	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.5	26.6	22.6	17.5	25.8	21.6	16.5	24.8	75.3	28.1	71.6	27.1	68.1	26.1	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	10.0	8.6	DB	-33.9	32.9	2.8	1.4	-35.9	33.9	-37.6	34.7	-39.2	35.5	-41.2	36.5	(4)
(5)			WB	-34.0	26.1	2.8	1.3	-36.0	27.0	-37.7	27.7	-39.2	28.5	-41.2	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-20.0	-14.7	-4.7	6.1	13.7	20.0	22.7	20.8	14.3	5.4	-6.7	-17.2	(6)	
	DBStd	15.52	4.29	5.26	6.19	4.33	3.89	3.22	2.26	2.65	3.58	4.41	6.43	4.79	(7)	
	HDD10.0	3727	930	693	456	133	10	0	0	0	7	154	502	843	(8)	
	HDD18.3	5758	1188	926	714	368	152	19	1	9	127	401	752	1101	(9)	
	CDD10.0	1315	0	0	0	15	124	301	392	335	137	11	0	0	(10)	
	CDD18.3	303	0	0	0	0	7	70	134	85	7	0	0	0	(11)	
	CDH23.3	2122	0	0	0	7	131	546	887	489	61	1	0	0	(12)	
	CDH26.7	507	0	0	0	1	27	158	231	86	5	0	0	0	(13)	
(14) Wind	WSAvg	3.5	3.7	3.9	4.2	4.2	4.0	3.4	2.9	2.7	2.6	3.4	3.8	3.7	(14)	
(15) Precipitation	PrecAvg	596	4	5	10	26	56	89	154	136	64	33	11	7	(15)	
	PrecMax	869	12	15	34	63	134	189	383	264	204	92	38	13	(16)	
	PrecMin	369	0	1	1	1	11	9	58	8	9	7	1	1	(17)	
	PrecStd	112	3	3	8	18	33	39	68	69	39	23	8	3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.8	3.2	13.3	24.0	29.5	32.3	32.4	30.3	27.4	21.9	12.4	1.4	(19)	
		MCWB	-7.3	-0.5	6.8	12.4	16.8	20.6	23.4	23.7	20.2	13.8	7.6	-0.2	(20)	
	2%	DB	-8.8	0.1	9.6	20.6	26.8	30.3	30.5	28.9	25.1	18.3	7.7	-4.3	(21)	
		MCWB	-9.8	-2.5	4.3	10.9	16.0	20.1	23.3	23.2	18.8	11.7	4.0	-5.5	(22)	
	5%	DB	-10.8	-3.0	6.7	17.7	24.4	28.5	29.2	27.5	23.2	15.3	4.5	-7.0	(23)	
		MCWB	-11.7	-4.7	2.3	9.4	14.9	19.8	23.1	22.3	17.4	9.9	1.9	-8.0	(24)	
	10%	DB	-12.6	-5.5	4.3	14.6	22.0	26.6	27.7	26.1	21.5	13.0	2.1	-9.5	(25)	
		MCWB	-13.4	-7.0	0.7	7.8	13.7	19.4	22.7	21.6	16.3	8.8	0.0	-10.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.0	-0.2	7.8	13.4	19.0	23.6	26.3	25.5	21.8	15.0	7.8	-0.6	(27)	
		MCDB	-5.9	2.7	12.0	20.7	26.0	29.0	29.6	28.8	25.3	20.7	12.1	1.8	(28)	
	2%	WB	-9.8	-2.4	4.7	11.5	17.2	22.2	24.9	24.2	20.1	12.6	4.3	-5.5	(29)	
		MCDB	-8.9	-0.1	8.8	19.5	23.8	27.3	28.3	27.4	23.6	16.4	7.1	-4.4	(30)	
	5%	WB	-11.6	-4.8	2.7	9.9	16.0	21.3	24.1	23.3	18.5	10.8	2.2	-8.0	(31)	
		MCDB	-10.9	-2.9	6.3	16.6	21.9	26.2	27.6	26.2	22.0	14.4	4.4	-6.9	(32)	
	10%	WB	-13.3	-7.1	1.0	8.3	14.7	20.3	23.3	22.4	17.2	9.0	0.1	-10.4	(33)	
		MCDB	-12.6	-5.6	3.9	13.9	20.7	24.8	26.6	25.2	20.3	12.7	1.9	-9.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	13.2	11.6	11.9	11.7	9.6	8.3	8.8	12.3	11.3	9.8	10.2	(35)	
		MCDBR	12.7	14.8	13.4	17.5	16.4	13.0	11.0	10.9	14.6	15.6	12.4	12.1	(36)	
		MCWBR	11.8	12.4	8.6	9.2	7.4	5.1	5.0	5.7	8.0	9.1	9.1	10.9	(37)	
	5% WB	MCDBR	12.6	14.6	12.3	15.8	13.3	10.4	8.9	9.0	12.0	13.8	11.9	12.2	(38)	
		MCWBR	11.7	12.3	8.1	8.8	6.5	4.7	5.0	5.3	7.2	8.9	9.0	11.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.354	0.432	0.534	0.576	0.589	0.561	0.500	0.451	0.458	0.383	0.338	(40)	
	taud		2.056	1.985	1.876	1.756	1.729	1.794	1.893	2.042	2.080	1.933	2.027	2.033	(41)	
	Ebn at Noon		728	786	772	728	714	708	723	753	752	658	646	657	(42)	
	Edh at Noon		100	131	170	211	224	211	189	157	138	138	103	92	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.07	3.20	4.26	4.69	5.22	5.52	5.05	4.69	4.12	2.86	1.90	1.62	(44)	
	RadStd		0.16	0.19	0.33	0.42	0.51	0.65	0.48	0.38	0.33	0.20	0.15	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+46	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page